

NIST-F2 is a caesium fountain atomic clock that, along with NIST-F1, serves as the United States' primary time and frequency standard. NIST-F2 was brought online on 3 April 2014.

Accuracy

NIST-F1, a cesium fountain atomic clock used since 1999, has a fractional inaccuracy ($\delta f / f$) of less than 5×10^{-16} .

Anticipated accuracy

The planned performance of NIST-F2 is $\delta f / f < 1 \times 10^{-16}$. At this planned performance level the NIST-F2 clock will not gain or lose a second in at least 300 million years.

Evaluated accuracy

The evaluated accuracy (uB) reports of various primary frequency and time standards are published online by the International Bureau of Weights and Measures (BIPM). The first in-house accuracy evaluation of NIST-F2 reported a uB of 1.1×10^{-16} . In March 2014 and March 2015 the NIST-F2 cesium fountain clock reported a uB of 1.5×10^{-16} in the BIPM reports of evaluation of primary frequency standards.

The last submission of NIST-F1 to BIPM TAI was February 2016.

Replicas

At the request of the Italian standards organization, NIST manufactured many duplicate components for a second version of NIST-F2, known as IT-CsF2 to be operated by the Istituto Nazionale di Ricerca Metrologica (INRiM), NIST's counterpart in Turin, Italy. As of February 2016 the IT-CsF2 cesium fountain clock started reporting a uB of 1.7×10^{-16} in the BIPM reports of evaluation of primary frequency standards.

References

External links

NIST Time and Frequency Div. — 2004: Strategic Focus 1 (National Institute of Standards and Technology)

"NIST Launches a New U.S. Time Standard: NIST-F2 Atomic Clock" on YouTube

"Time gets an upgrade". newscientist.com. New Scientist. Retrieved 27 February 2018.